

ALIAH UNIVERSITY

End Semester Examination (Spring Semester) 2026 (3rd Year 2nd Semester B.Tech Program)

Subject Name: Compiler Designing

Full Marks: 80

Time: 3 Hours

Subject Code: CSAUGPC32

Group A

Answer all 5 questions

5 X 2 = 10

1. Answer the following in at most two sentences

- a) What is a compiler? (CO1 - Remember)
- b) Define lexical analysis. (CO1 - Understand)
- c) What is a parse tree? (CO2 - Understand)
- d) What is intermediate code? (CO3 - Understand)
- e) Define symbol table. (CO2 - Understand)

d) compiler is a system or machine to analyse the code and ~~convert~~ ^{check for} errors if no error found it **Group-B** converts the source code into machine code.

Group-B

Answer any 6 questions

b) Lexical Analysis is the phase where source code are **6 X 5 = 30** converted into tokens.

2. Explain the phases of a compiler with a diagram. (CO1 - Understand)

3. Describe the role of lexical analyzer. (CO1 - Understand)

c) Free line structure
that explains

4. Explain top-down parsing with an example. (CO2 - Understand)

5. Discuss bottom-up parsing techniques. (CO2 - Understand)

6. Construct DFA for a given regular expression. (CO3 - Apply)

7. Explain error handling techniques in compiler design. (CO4 - Understand)

8. Describe syntax-directed translation. (CO3 - Understand)

Group- C

Answer any 4 questions

4 X 10 = 40

9. a) Convert a regular expression to NFA- $r=(a|b)^*abb$ (CO3 - Apply)

b) Convert the NFA to DFA. (CO3 - Apply) 5+5

10. a) Compute FIRST and FOLLOW sets for all non-terminals.

$S \rightarrow Aa|bA$

$A \rightarrow c|e$

(CO2 - Analyze)

b) Construct the LL (1) parsing table and check whether the grammar is LL (1). (CO2 - Analyze) 5+5

11. a) Construct the LR (0) items for the grammar.

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid id$

(CO2 - Analyze)

b) Build the SLR parsing table for the grammar. (CO3 - Apply) 5+5

12. a) Generate intermediate code for arithmetic expressions $x = a + b * (c - d) / e$ (CO3 - Apply)

b) Represent the above expression using: Quadruples and Triples (CO3 - Understand) 5+5

13. a) Explain code optimization techniques. (CO4 - Understand)

b) Discuss loop optimization methods. (CO4 - Analyze) 5+5

Intermediate code:-

$$x = a + b * (c - d) / e$$

$$t_1 = (c - d)$$

$$t_2 = t_1 / e$$

$$t_3 = a + b$$

$$t_4 = t_3 * t_2$$

$$x = t_4$$

Tripple:-

operator	sch 1	sch 2
-	c	d
/	t(1)	e
+	a	b
*	t(3)	t(2)
=	(5)	(5)

Quadruple:-

	operator	sch 1	sch 2	Result
1)	-	c	d	t ₁
2)	/	t ₁	e	t ₂
3)	+	a	b	t ₃
4)	*	t ₃	t ₂	t ₄
5)	=	t ₄		x